J.A.R.V.I.S. — OS

Voice-first edge AI operating system for smart glasses · Funding request (India · INR)

Submission target: 10 November 2026 | Proposal v5.0 (India edition) | Prices verified 5 November 2026

1. WHAT ALREADY EXISTS (NOT A CONCEPT)

JARVIS-OS is a working software stack, not a slide deck. The following already runs on a standard Windows/Linux host and is covered by 24 automated tests:

• **Voice pipeline:** 16 kHz dual-engine microphone capture, offline-capable speech-to-text, neural text-to-speech in four distinct agent voices (JARVIS / FRIDAY / EDITH / ULTRON).

• **Brain:** OAuth-authenticated model routing (zero API-key handling, zero cloud spend), with voice-driven switching between models and a deterministic intent layer for common commands.

• **Tool layer:** permission-gated host automation — launch applications, screenshot, volume, system telemetry, outbound telephony dispatch — with an append-only action log.

• **Shell:** bootable image plus a kiosk HUD (glass panels, on-demand 3D holograms, real-time webcam gesture tracking with 12 gestures).

The grant is requested for **hardware only** — to move this proven stack from a desktop prototype onto wearable, field-testable hardware.

Technology readiness, stated honestly. The desk-bound software stack is at roughly **TRL 6** (working system demonstrated in a relevant environment). The **wearable product does not exist yet** — no glasses-mounted build, no on-device inference, no field trial. That wearable prototype is at **TRL 3**, which is exactly the stage these schemes exist to fund. This proposal asks for prototype hardware, not for money to finish software that is already finished.

2. ITEMISED BUDGET (INR, GST-INCLUSIVE ESTIMATES)

#	Component	Role in the system	Cost (■)
1	**Google Pixel 10a** **8 GB / 256 GB**	Pocket inference host. Tensor G4 NPU runs the quantised on-device model so the assistant works fully offline with zero per-query cost.	49,999
2	**Ray-Ban Meta (Gen 2)** **Wayfarer, clear lenses**	Wearable camera + open-ear audio + dual microphones. The actual field-test target for the glasses interface.	42,100
3	**NVIDIA Jetson Orin Nano Super** **8 GB dev kit, 67 TOPS**	Edge vision board. Sustains 30 FPS hand-gesture tracking and hologram rendering without a desktop GPU.	42,000
4	**Quectel EC200U 4G Cat-1 modem** **+ antenna, enclosure, SIM**	Standalone cellular path for real voice calls independent of a paired phone.	3,500
5	**WebRTC TURN relay** **India region VPS, 12 months**	Full-duplex two-way audio between glasses, phone and host when NAT traversal fails.	12,000
6	**Power & field rig** **24,000 mAh PD 65 W, cabling, heatsinks**	Continuous wearable operation for a full demo day.	6,500
	Sub-total		**1,56,099**
7	**Contingency 10%**	Price movement, import duty, replacement of a failed unit during field trials.	**15,610**
	TOTAL REQUESTED	**≈ US$ 1,960**	**■1,71,709**

Indicative India retail prices captured on 5 November 2026 (Google Store India, Amazon.in, Robu.in, Indian Jetson distributors). Final invoices will be filed with the incubator; unspent contingency is returned.

3. WHICH INDIA SCHEME THIS FITS

Route	Ceiling	Status as of 5 Nov 2026	Fit for this project
NIDHI-PRAYAS 2.0 **DST, Govt. of India**	■20 L (PRAYAS Centre) ■40 L (Advance PC)	Open — rolling calls run by each PRAYAS Centre. No application fee, no equity taken.	**Primary. Priority areas name "Hardware-based Artificial Intelligence and its application to Indian problems" and "assistive devices". An individual innovator may apply without incorporating a company first.**
MeitY TIDE 2.0 **MeitY Startup Hub**	■7 L EiR grant ■25 L seed	Open — delivered through 51 approved incubators; several are inviting applications now.	**Fallback in the same cycle. Same prototype ask, but the seed tranche expects an incorporated entity.**
TANSEED 9.0 **StartupTN (Tamil Nadu)**	■10 L / ■15 L	Closed for 2026 (round 8 shut Jan 2026). Next round expected Dec 2026.	Year-end follow-up, and only for a Tamil Nadu registered entity with DPIIT recognition.
Startup India Seed Fund (SISFS)	■20 L grant + ■50 L debt	New applications closed 31 May 2026; existing grantees continue to be disbursed.	Not usable for a 2026 submission. Listed so no reviewer assumes it was overlooked.

NIDHI-PRAYAS explicitly does **not** fund pure software, and does not fund products that have already crossed MVP. This application therefore covers only the **wearable hardware prototype** (glasses mount, on-device inference board, cellular modem, power rig) — the software it runs is already written and is contributed by the applicant at no cost to the grant.

4. WHY THIS DESERVES THE MONEY

• **Indian-problem framing.** The same stack that controls a workshop desktop can give a visually impaired or motor-impaired user hands-free control of a phone, and can run fully offline in low-connectivity districts — voice in, voice out, no cloud bill.
• **Zero recurring cost.** On-device inference on donated hardware means no subscription or API spend after the grant is closed, which is what most public-funding reviewers ask first.
• **Auditable safety.** Every tool call is allow-listed and logged; destructive actions require explicit spoken confirmation. This is a design requirement, not a feature.
• **Small, defensible ask.** ■1.72 lakh buys the complete field kit. This is a prototype-validation request, not a payroll request.

5. MILESTONES TO 10 NOVEMBER 2026

Date	Deliverable	Evidence produced
5 Nov (done)	Working voice stack, 24/24 tests green, bootable image built.	Test log + commit history.
6–7 Nov	On-device model deployed and benchmarked on the target phone class.	Offline latency and token-rate table.
8 Nov	Full-duplex WebRTC call path, two-way audio verified.	Recorded call, < 200 ms one-way latency.
9 Nov	12-gesture recognition tuned under indoor and outdoor light.	30 FPS capture with per-gesture accuracy.
10 Nov	**Live demonstration + submission pack.**	**Grant reviewed against a running system.**

Applicant: JARVIS-OS project · **Entity type:** individual innovator / early-stage startup (DPIIT recognition in progress)
Seeking: ■1,71,709 non-dilutive hardware grant · **Deliverable:** field-tested wearable AI assistant, open documentation, and test suite
Contact channel: via the incubator / PRAYAS Centre application portal